From a single number to a landscape: topology-resolved quantum volume

Quantum Science and Technologies

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Goal of the presentation

Structure

1. NISQ computers,
2. Noisy simulators,
3. Quantum volume,
4. Research demo

NISQ computers

Noisy simulators

Quantum volume (1)

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Heavy set of outputs

Let $\mathbf{Q}$ be a random circuit drawn from a suitable ensemble acting on n qubits. The quantum state after executing the circuit is denoted $|\psi\rangle$. Each possible output state $x \in \{0,1\}^n$ is measured with probability $|\langle x|\psi\rangle|^2$. The set of output states with a probability greater than the median constitutes the heavy set of outputs associated with the quantum circuit $\mathbf{Q}$.

Quantum volume (2)

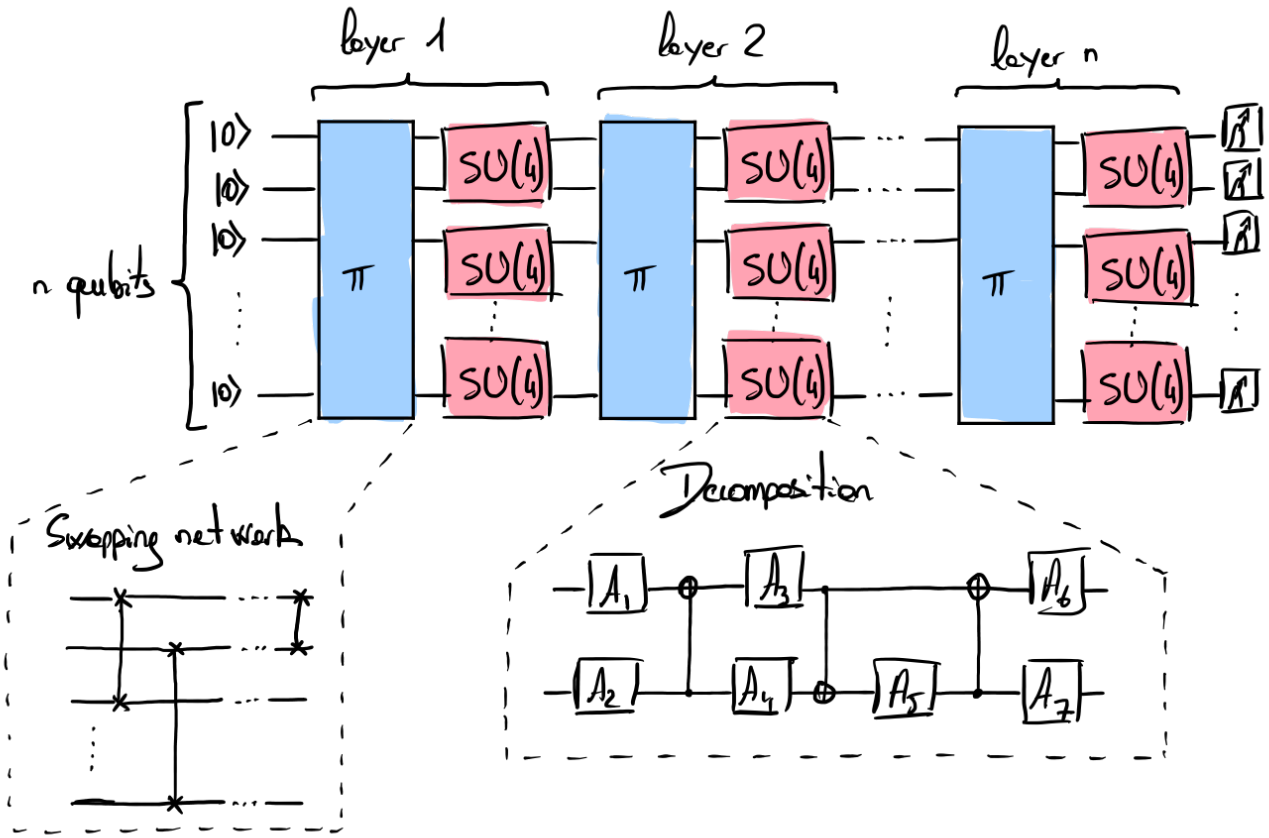

Figure: The QV circuit. Source: `https://quantumbenchmarkzoo.org/content/system-level-benchmark/others/quantum-volume`

Quantum volume (3)

If the quantum computer samples a distribution from the **heavy set of outputs** (simulated classically) with probability $> 2/3$, it validates the corresponding quantum volume score of 2^n. The confidence interval for this evaluation is set to 2σ (97.7%).

Problem

Quantum volume assesses only a limited subset of best-quality qubits within a quantum processor and does not provide a comprehensive measure of the overall fidelity or performance of quantum operations across the entire chip.

Research methodology

We benchmarked QV on the IBM Marrakesh device using calibration data and the coupling graph from Qiskit.

1. Construct a noisy quantum simulator from the extracted data
2. For each size $n \leq 6$:
    1. Enumerate all connected subgraphs of size n
    2. Run QV circuits on each subgraph
    3. Compute heavy output probability and confidence

Results

n	Count	$\langle HOP \rangle$	$\sigma(HOP)$	Median	Min	Max	$\langle SE \rangle$	Lower
3	244	0.664	0.063	0.688	0.438	0.712	0.047	0.572
4	354	0.732	0.083	0.766	0.503	0.817	0.043	0.647
5	566	0.733	0.085	0.769	0.498	0.825	0.043	0.648
6	926	0.748	0.085	0.779	0.491	0.832	0.042	0.665

Discussion

For QV, each layer is built from random SU(4) two-qubit blocks after a random permutation, and for odd widths one qubit is idle each layer. On a connected 3-qubit path like (a, b, c), one of the possible two-qubit interactions is between the two endpoints. That interaction is not native, so with trivial layout and basic routing it tends to introduce SWAP overheadresulting in extra routing cost that can dominate the whole layer. 3Q layers contain only one gate, so routing dominates. 4Q layers contain two gates, allowing routing overhead to be shared.

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